

The Sex Peptide in *Drosophila melanogaster*

melSP is a component of the seminal fluid that induces post mating responses in *Drosophila* females. It is gradually released by proteolytic cleavage and induces several long-term postmating responses. What does this entail? SP acts on internal sensory neurons expressing *pickpocket*(ppk). These then induce PMR.

Classic Post Mating responses

- elevated egg-laying
- reluctance to further mating
- altered dietary preference (high protein:carbohydrate interference) - (testing nutrient deficient flies by activating this group of neurons?)
- changes in sleep patterns
- JH biosynthesis - and immunosuppression
- Female aggression

Females that lack SPR continue to show virgin like behaviours after mating. SP is thought to pass from the reproductive tract into the haemolymph and then directly into the brain and the CNS. Post-mating responses can be induced in virgin females not only by injection of SP but also by blocking synaptic transmission of neurons that express the sex-specific transcripts of the *fru* gene.

Neuronal circuitry for PMR in *Drosophila* females

SP mediates PMR through sensory neurons that co-express the *fruitless* gene and *pickpocket* gene in female reproductive system. How these signals are shared to the central circuitry remains unknown. (?)

Important circuits:

1. *dsx* circuitry associated with *fru*⁺/*ppk*⁺-neurons
In *dsxGal4* females expressing TNT no egg deposition is ever observed, with unfertilized eggs atrophying in the lateral oviducts. In contrast, when *fru*⁺ neurons are silenced, deposition of successfully fertilized eggs is still observed, suggesting that different subsets of the *dsx*⁺/*fru*⁺/*ppk*⁺ SP-responsive sensory circuit may direct distinct postmating behavioral responses.
2. Reproduction profoundly changes nutritional requirements, with many species showing an appetite for sodium during reproductive periods. How this internal state modifies neuronal information processing to ensure homeostasis is not understood. This study shows that dietary sodium levels positively affect reproductive output in *Drosophila melanogaster*; to satisfy this requirement, females develop a strong, specific appetite for sodium following mating.
3. Studies have demonstrated that SP has an additional, unsuspected role in males in the assembly of neutral lipid-containing microcarriers in the AG lumen. These microcarriers store SP and can carry other proteins with lipid anchors. Furthermore, proteomics analysis reveals that the normal delivery of subgroups of SFPs to females during mating requires SP, potentially because these subgroups interact differently with microcarriers. Microcarrier interactions are likely to also affect dispersal of these proteins in the female reproductive tract. This analysis of microcarriers in other *Drosophila* species reveals that SP's microcarrier assembly function may exist in species in which SP has more limited roles in modulating the PMR, suggesting that the former function might have been critical in the evolution of this molecule.
4. Identification of extracellular neutral lipid microcarriers as accessible stores of specific seminal proteins is reminiscent of the role of intracellular lipid droplets in storage of cytoplasmic and nuclear proteins. Lipid droplets are able to dock with specific intracellular organelles to mediate

their functions and deliver their cargos. It will be interesting to investigate whether the remnants of microcarriers, such as the microdomains observed with SP-GFP, are in any way targeted to specific cells or structures after transfer to females as these storage vehicles break down.

5. How do males do this? Based on cVA?

It has previously been reported in *Drosophila* that males can adaptively modulate the relative balance of seminal proteins, including SP, in the ejaculate, depending on female mating status and the presence of rival males. Loading of selected proteins onto microcarriers might provide a simple mechanism to control such rapid changes if the transfer of these large structures can be differentially regulated compared to soluble proteins: for example, by controlling the opening of the sphincters through which seminal fluid passes from the AGs to the ejaculatory duct.

Investigating whether any of the network genes is involved in loading or unloading SP from microcarriers or, indeed, whether they play a role in microcarrier assembly, particularly since they appear to be present in species where SP does not seem to be involved in signaling. The role of secondary cells in microcarrier morphology also needs to be examined in more detail.

Furthermore, confirming that other SFPs identified in the proteomics analysis or main cell-expressed GPI-anchored proteins are microcarrier cargos should allow the functions of these structures to be assessed more extensively and may suggest molecular tools that could be used to screen for similar processes in higher organisms